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Review on Medical Image Segmentation and Feature Extraction Approaches

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***Abstract*—** In medicine, "imaging" refers to the technique or technology of seeing in or out of a living organism to discern the constituent parts of that body. During the operation, the medical diagnosis is examined, an analysis of the disease is performed, and image data sets that are considered normal and abnormal are created. Two different types of images are used in the medical profession. These are medical images that use visible light and those that use invisible light. A radiologist can only decipher the first category of medical images, but a person of average intelligence can understand the second category of medical images. Both segmentation and feature extraction are required to analyze all of them. The authors of this study limit the scope of their research to tumor detection by mammography or magnetic resonance imaging. While there is a wide range of medical imaging techniques, they focus on these two approaches. The authors of this article address several methods for the medical image preprocessing step, known as segmentation and feature extraction.

***Index Terms*—** *Medical Imaging, Obsolete Segmentation, Recent Segmentation, Feature Extraction.*

I. INTRODUCTION

Image processing techniques are becoming increasingly significant in many applications, which need increasingly complex tools and methods. Image segmentation is a prime example of a topic that falls under the umbrella of image processing, as well as another essential component and the primary focus of image processing techniques [1]. Image segmentation is a method that involves splitting an image into sub-component sections that are astonishingly similar to one another in terms of their properties. This procedure enables you to extract relevant information from an image. The process of Image segmentation holds an important place in the field of image analysis. The accuracy of a medical Image has a role in patient care since it affects both the diagnosis and the therapy that may be administered [2]. In medical image analysis, segmentation plays a significant role and seeks to accomplish the goal of extracting certain features from the Images [3]. These kinds of Images can be utilized for more advanced visual comprehension. Segmentation is a hypothetical mid-level vision job done by neurons between low-level and high-level cortical regions, as represented schematically in Figure 1 [4].

According to the scientific community, segmentation is an essential part of visual perception. Clinical Image segmentation aims to identify anatomical structures and depict the boundaries of those structures using a digital source. Images are a crucial component of the treatment process, particularly in radiotherapy (RT), because they recognize the therapy target and ordinary structures to prevent irradiation [5]. Therefore, Radiation treatment planning systems (RTPS) require anatomical data to be displayed on Computed tomography (CT) scans to function correctly. On CT scans, medical professionals will physically display the treatment aim and the typical facilities. In order to determine the appropriate radiation dose, segmented clinical Images are sent to a radiation treatment planning system [6].

As a result, proper segmentation is essential for the patient's successful treatment. For administering radiation treatment, the segmentation quality considers both the spatial precession and the precision in dosage computation, two strongly associated factors [7]. The goal of image segmentation is to break down a medical Image into components or things that are fundamental to the image and have consistent and standard features. Intensity is one of them. It is associated with anatomical structures, cancers, and other things. The application determines the level of segmentation required, and no one theory can be applied to all cases of Image segmentation [8]. The research community has developed many different segmentation strategies or algorithms throughout the years. These strategies circumvent several limitations that are inherent in traditional medical segmentation methods.

The nature of the problem and the type of Image being processed both play a role in determining whether specific approaches or algorithms should be prioritized above others [9].

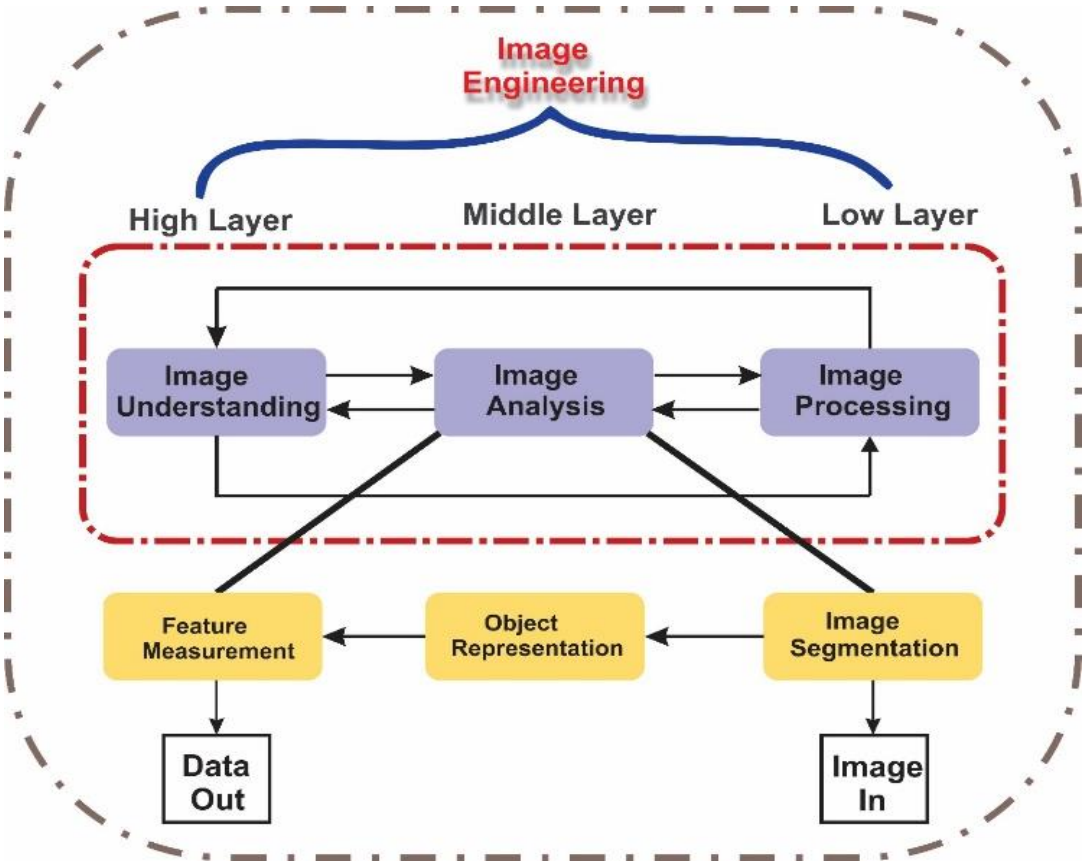


Fig 1 Image Engineering and Segmentation

II. LITERATURE REVIEW

Initially, Convolutional neural network CNNs such as Google Net, AlexNet, VGC, and ResNet were examined to see how well they performed image classification tasks [10]. Previous research has already altered these networks and improved their pixel categorization. Thus, this has been done. Labeling pixels using a patch or region-based approach is easier with these nets' assistance.

The old approaches are less accurate than the networks that have been developed. On the other hand, these networks have a few drawbacks, which become apparent throughout the testing process. For the fully convolutional network to function, it requires an entire Image to serve as its input. The network then generates the output in the form of a dense segmentation. In addition, there is an issue with the loss of the spatial pattern since there was more than one maximum pooling phase, and down sampling was used. This approach samples the layers using the deconvolution stack and the activation function [11], which allows it to circumvent the disadvantages described above. These aspects of the convolutional network and the deconvolutional framework motivated the development of a U-shaped network named U-Net. The segmentation of biological Images was the primary application of the three-dimensional version of this U-Net architecture. A similar kind of architecture known as V-Net for prostate Magnetic resonance imaging (MRI) image segmentation has been suggested as a viable option. Both successive down sampling and up sampling have their advantages and disadvantages [12]. The resolution of the feature map has suffered to some degree. It has been suggested that such a dilated convolution would protect the architectural characteristics of the feature maps and, as a result, enhance the capacity of the receptive field to take in more comprehensive textual information. Object tracking and semantic segmentation are processes in a dilated convolution heap. Extended convolution is not only utilized for detection segmentation but also for detecting the activities of video graphics frames [13]. This is only one of its many applications.

The multi-scale functionality may also be extracted with the help of the convolution network. This is done in order to increase the precision of the targeting. Multi-scale capabilities may be accomplished by sending various

forms of the input Image through the same network. Pixels may then be classified based on the properties gathered in this manner. The features of each pixel are derived from two homocentric patches of varying sizes, and these patches are used to create the image. When dealing with multi-scale Images, these are sent through many steps before being processed by a recurrent convolutive network. Utilizing feature maps that consider disproportionate levels of convolutional neural networks is yet another common application of multi-scale functionality. For the goals of both localization and segmentation, the properties of the intermediate layers are strengthened [14].

Such a method of widespread use of image segmentation is known as interactive image segmentation and may be applied in various contexts. There are several other ways for users to interact with the system, such as clicking, outlining, and using bounding boxes. In addition to Graph Cuts, Random Walks, and Geodesic object segmentation (GeoS), you may also create scribbles. On the other hand, the majority of them are utilized in lower-level operations, which call for additional user flags. Images with a more profound outline or gloomy edges are more accessible to control due to this. Some technologies are presented for machine learning that can learn from users' bookmarks. These algorithms are not capable of improving dissection accuracy while requiring less input from the user. However, they can only rely on the limited manual functions they have acquired from their own experience. The level of interactive segmentation was improved using a deep convolutional network. The convolutional neural network demonstrates not only outstanding performance but also possesses spontaneous feature learning. Some examples include a three-dimensional U-Net that acquires knowledge from the labeled Images and is subsequently used for a segmentation that is not automated [15]. In addition, the doodles are utilized in the training of CNNs. When S-segmentation of MRI images is required, Deep Cut trains the convolutional neural network using the user's specified bounding box as markers, this is done to meet the requirements. Some processes are not fully communicative and not even suited for testing because all they do is accept repeated improvement suggestions. These processes should be avoided. When it comes to tools for the in-depth selection of communicative objects, tools have been developed in which the acquired Images are first transformed into Euclidean distance maps and then aggregated with the input of totally convolutional neural networks [16]. On the other hand, geodesic distance changes and encodes spatial dimensions while counteracting sensitivity; nevertheless, this technique has not been included in CNN's. Graphical models, such as conditional reference fields, are frequently used to raise the dissection precision by using spatial accuracies. This is done in several different contexts. If spatial regularization is performed, the minimal flow max/cutoff algorithm is utilized to achieve the goal of minimizing Potts energy. In order to give optimization formulas for the maximum flow issue, a mapping must first be done [17]. They ensure that the pixel segmentation of nearby pairs with comparable properties is carried out consistently. A wholly connected Cardiovascular research foundation (CRF) is utilized throughout the procedure of including long-distance connections in the Image segmentation process. This demonstrates that there is a possibility of a match between all of the Image graphs. A linear combination of Gaussian kernels is used to determine the capabilities of neighboring edges. Other solutions include built-in vector machines to support output and gradient-based optimization to improve performance [18]. These are also put to use in the process of learning CRF settings. After successfully segmenting the image, the next step will be extracting the features from it. Several different deep learning methods are now available for use in the process of functionality extraction from medical Images. Extracting features may be accomplished with the help of a convolutional neural network. Creating a model that can extract text features from medical Images and perform the analysis using a convolutional neural network is the focus of a significant amount of research into the feature extraction of medical data [19]. Following the completion of the last stage of image processing, known as feature extraction, we can utilize a classifier to examine the features based on their present abnormalities. When classifying the processed Images, we can utilize either a convolutional neural network or a more conventional classification approach. Image recognition may be accomplished with the help of a wide variety of advanced algorithmic approaches. It is well known that the Support Vector Machine method is one of the most prominent classification algorithms utilized in various medical image processing and analysis procedures [20].

III. MEDICAL IMAGE SEGMENTATION APPROACHES

Several scholars have presented different segmentation methods, however none of these researchers have yet proposed a universal approach that can be utilized for various purposes [21]. An Image can be segmented into several sections according to its characteristics. Each zone has the same characteristics in terms of brightness, color, texture, and responsiveness. In general, segmentation is identifying sections of interest to work on or support the description of anatomical structures. The authors focus primarily on mammograms and the image segmentation techniques used in this section. The identification of masses, micro calcifications, and speculative lesions is facilitated due to this [22]. In addition, it assists in approximating the breast density based on the segmentation of dense tissue areas. The following illustration, titled "Image Segmentation Practices," which may be found in the literature, is referred to as Figure 2. In light of this review, the writers focus on introductory information on the methods above [23].

On the other hand, one may get complete information on these methods by looking at the references that go along with them. Following the examination, it has been abundantly evident via further research and analysis that the segmentation process in medical imaging may be roughly divided into two groups: obsolete techniques, and modern procedures. Even though they are considered antiquated compared to more modern practices, the old methods are nonetheless put into practice today. In the following paragraphs, the writers will go through the fundamental ideas related to the procedures [24].

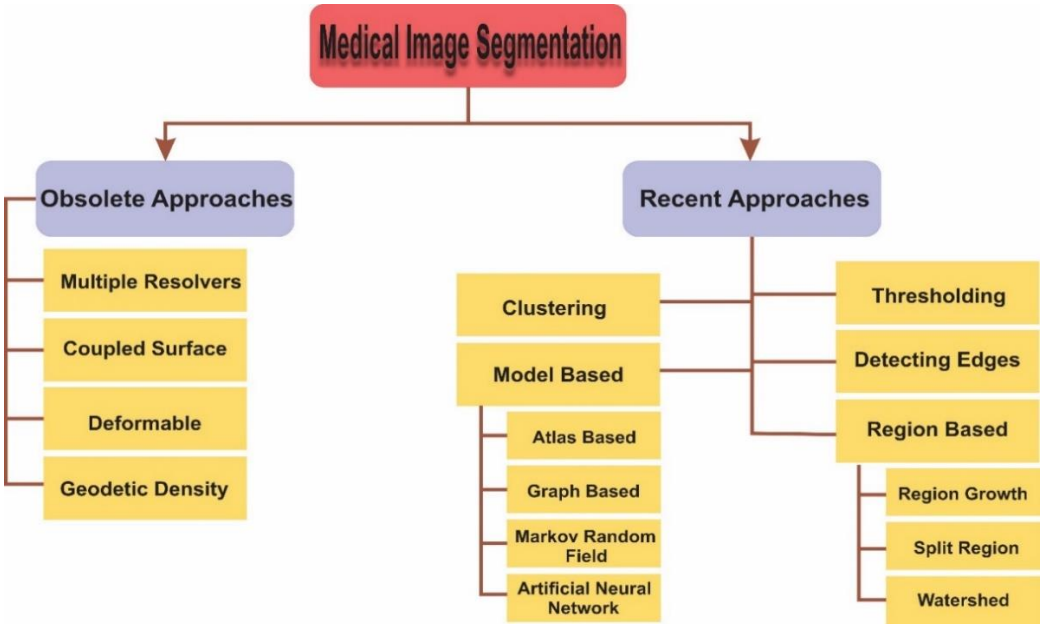


Fig 2 Several Image Segmentation Approaches

A. Obsolete Segmentation Approaches

Techniques of segmentation that were established in the late 1990s but are now largely defunct were created then. The deformable model's approach, the multiple resolution methods, the linked surface method, and minimum geodetic mathematics are all examples of segmentation techniques that come under this category [25]. This section provides a concise discussion of these various methods.

1) Methods with Multiple Resolvers: The Multiple Resolver approach provides a resolver of the highest caliber and is adaptable to various error conditions and data. By looking at the Images of each piece, we can see that there are complex contracts associated with structures of a specific spatial scale. In order to fulfill the scale of attention, the extension of the standard components of the image needs to be given permission. This may be accomplished by employing a generic segmentation approach predicated on the similarity of meanings and combining it with both local and global optimization [26-27]. In addition, it discusses a technique for obtaining an accurate approximation of the edge orientation by calculating the root mean square error at several resolutions. In order to provide a more accurate edge location and orientation, this method makes compelling advantage of the spatial constancy offered by minor kernel gradient operators operating at varying scales. The primary benefit of using it is that this approach can extract the border orientations after the data while maintaining a high signal-to-noise ratio [28].

2) The Coupled Surface: In many different medical Image scans, the issue of correctly segmenting a volume layer with a restricted width is tackled. The border facades and the analogous environments created by the gray level standards between the facades continue to exist as a layer. The final description of the boundary facades and the involuntary segmentation of the layer are obtained using two embryonic facades that are instantly entrenched. Each of these embryonic facades is guided by specific information derived from an image of the situation while the coupling is maintained. A local operator who bases their decisions on gray level values is considered to receive knowledge regarding the surface verdict of ambition. When information based on gray levels rather than the same gradient is used, the apprehension of homogeneity is enhanced, encouraging enactment [29].

3) Deformable Models: The boundaries of things can be described using deformable models in terms of their exterior curvatures or deformations. In the field of medical imaging, it has found widespread application in the process of Image segmentation, particularly for the segmentation of anatomical features. To accomplish this, basic information on the item's geometry is incorporated into the modeling process, and the edge produced is constrained to be continuous and smooth. During deformation, the curve can be smoothed out by internal

forces that originate from the surface or the curve itself. External services do calculations using the Image data to alter the curve or surface up to the object's border [30].

4) Geodetic Density Method: The geodetic grade method, also known as the geodetic density method, is one of the Image segmentation approaches utilized more frequently. The geodesics between an assumed base argument and a collection of arguments spread throughout the aspect may be computed using this approach. The number of geodesics that ignore a particular pixel provides a clear indication of the density of geodesics in each pixel of the aspect. Obtaining access to spawn places with a high geodesic density is the responsibility of the destination organization. In most cases, the client is responsible for providing the start and finish locations of the Image, and in return, they receive the shortest path possible. These minimum pathways are identical to geodesics that have minimal length. Find a collection of curves that, when applied worldwide, result in the minimum energy consumed by active geodetic borders. In addition, several requirements must be met to automatically retrieve a collection of endpoints by supplying just one beginning point [31].

B. Recent Segmentation Approaches

Recently, a significant amount of attention has been paid to the segmentation process. The research that has been done on segmentation has looked at several different approaches and algorithms. These strategies circumvent several inherent limitations in more traditional approaches to medical division. However, no one method can be used to determine which approach is the best for the many Images; these methods are only appropriate for the applications and images they are designed for threshold, region growth, region merging and division, clustering, edge detection, and model-based algorithms are the several ways image splitting techniques may be categorized [32]. The intensity levels of the data and its discontinuity and similarity are the two primary factors upon which all segmentation algorithms are predicated. The Image is then segmented based on an instantaneous shift in either the intensity levels of an image or the gray levels of an image by the discontinuities. Within the context of this method, the identification of singular points is the primary focus of our attention. The second strategy is based on pixels comparable in particular ranges according to pre-set criteria used to segment Images. This approach includes methods such as Threshold, Region Growth, and Divide and Merge Regions. Images are segmented according to these pre-set criteria [33].

1) ***Clustering Approach:*** Clustering groups homogenous data by similarity. "K-means clustering (hard clustering)" is a cluster segmentation procedure where dataset components belong to just one cluster. The soft clustering approach named as fuzzy c-mean (FCM) allows a pixel to be in more than one cluster. The fuzzy community function value between 0 and 1 determines where to place the pixel. The pixel is near the center of gravity and may be limited if the community function is 1. The goal function is the sum of squared Euclidean separation between each input sample and the cluster center [34]. The FCM algorithm divides grayscale and color Images into predetermined categories. Differences from projected values might reveal segmentation flaws. FCM can balance and expand the objective function. FCM is versatile because of its variable objective function. Minimizing computing effort gives acceptable results since the FCM method measures validity. FCM algorithm improvement was underway. When accounting for MR image intensity in homogeneities, FCM algorithms are noise-sensitive. Kernelized fuzzy C-means replace Euclidean separation [35]. The objective function has a dimensional penalty. He developed the fuzzy generalized fast clustering c-means (FGFCM) technique to suppress noise and maintain Image detail [30].

2) Model-Based Algorithms: Model-based techniques are the best for assessing model Images. This model shows shape and structure. This approach resists visual artifacts better than others.

Atlas-Based Model: Atlas uses Images and anatomical features to integrate earlier data for partitions and modify outcomes. An atlas uses clinically impaired and normal persons' indexes. Atlas can be physically labeled; it is related to the dividing table. Many atlases improve partition precession by registering Images [36]. The physically tagged Atlas was modified to "LABEL PROPAGATION" during the Image inscription to split the marked item precisely. Errors can arise if the Atlas and the object of study differ topographically [5]. Atlas developers choose source Images based on the actual population mean or the sample closest to the mean. Depending on settings, iterative methods can reduce atlas distortion. Due to the many atlas partitions, an extensive database is needed, and a query Image atlas must be determined. The atlas labels for mapping were limited by modifying the casting weights from the enrollment neighborhood evaluation [37].

Graph Based Methodology: The use of graph-based methods significantly aids the process of picture segmentation. In this method, a weighted graph denoted by the equation G = (V,E) is constructed by combining a collection of nodes denoted by V and a set of edges denoted by E. The network's vertices are the nodes, and the edges are the connections between the vertices. The arcs (vi,vj) E link the nodes vi,vj in an order determined by a neighborhood system. The nodes v ∈V correspond to the pixels that make up the image [38]. The cost associated with each node and arc (vi,vj) E conveys a particular level of preference that the corresponding pixels belong to the item of interest. The graph slicer resolves a region-based segmentation problem by applying combinatorial maximum and minimum flow optimization techniques developed by st. In addition to the node set V, it has two unique terminal nodes, denoted by the symbols source s and sink t. These terminals have a close relationship with

the segmentation seed points and stand in for the segmentation labels (object, background). The purpose of the graph section is to reduce the value of the objective function shown in the equation as much as possible [39].

$$C(L) = \lambda R(L) + B(L)$$

Here λ is a combination with weights of a property term called regional *R(L)* and a cut-off term called *B(L).*

Markov Random Field Models: The Markov random field (MRF) is a method for the probabilistic tagging of pixels used in the picture segmentation process. The contributed image is used as a source for the pixel label segmentation, extracting features. A feature vector is assigned to each pixel that makes up the picture. A string of tags works toward the completion of the picture. A label is distributed over the respective pixels in the image. In actual photographs, the regions are often consistent, and the pixels that are close to one another have, for the most part, comparable characteristics, such as intensity, color, and texture. A MRF stops are a suitable example of restrictions [40-41].

The random Markov fields are generally an objective mix of fundamental components such as the reflection field and the hidden labeling field, the pixels and their neighbors, the cracks and potentials of the cracks, the energy function, and the Gibbs distribution. In addition, the random Markov fields can be thought of as a mixture of the reflection field and the hidden labeling field. If every pair of pixels in a subcategory is pretty near one another, we refer to that subcategory as a crack. Figure 3 provides a graphical depiction of the information [42].

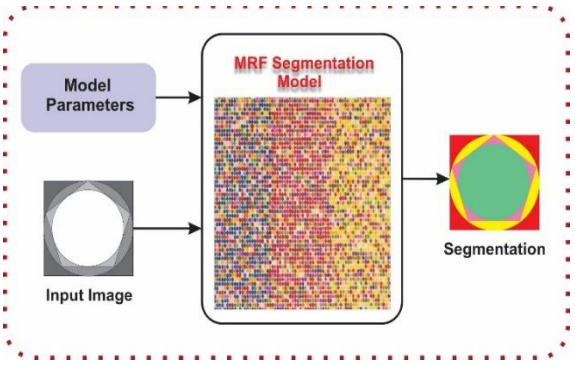


Fig 3 Graphical Representation of MRF Segmentation

Artificial Neural Networks: Inspired by biological neural networks, it is a mathematical model of neurons. The node is a copy of neurons with distinct functional components. Synaptic weights connect these nodes. These synaptic weights use the activation function to categorize or identify objects [43]. Neural networks learn and train. The neural networks are prepared with properties such as mean, standard deviation, kurtosis, asymmetry, or transfer using Wavelet or Curvelet transforms. During neural network training, the speculation phase lasts until the study state. The better the neural network's training, the better the test image. Learning is a customizable activity in which neuron weights are modified to generate feedback. Learning uses supervised and unstructured neural networks [44]. Neural network algorithms have trouble defining network design, size, type, number of layers, and geographic areas. These components affect problem treatment exposure. "Group data management method (GMDH)" identifies lungs from clinical Images. Fuzzy neural networks generate noise-insensitive segmentation [45]. He introduced SVM separation measures and studied hyperparameter selection. This approach provides high segmentation accuracy and needs minimal processing time and memory. He segmented hepatic neoplasms from M.R. images using a "3D fast walk algorithm" and "single hidden layer" feedforward neural network. The results are compared to a radiologist's physical description. Semiautomatic segmentation strategies decreased time unpredictably and produced accurate results.

3) Threshold Method: Threshold transforms a grayscale image into a binary image using a threshold value. This algorithm chooses the threshold value (T), foreground pixels with an intensity above the threshold, and background pixels. "The method of maximum entropy" and the Otsu approach's (maximum) variance are well-known industrial methods. K- means can be used similarly. Noise and dark edges impact threshold segmentation for Images with hard edges. Typical solutions include noise attenuation and edge threshold detection [46].

4) Detecting Edges: This is the most conventional Image flaw detection approach. Edge is the separation between two sections with different gray levels. Any program that enhances images will show edges. Image borders are created using derived operations and a convolution function with a mask [47]. Canny is a good edge detector that employs the gradient extension threshold to find the edge. Non-maximal suppression and hysteresis threshold repress. Noise in the Image impacts edge recognition, resulting in incorrect or jagged edges consisting of discrete pixels. Before edge detection, the image must be smoothed using the Gaussian operator. Image preprocessing can cause false and override edge detection at several resolutions [48].

5) Region-Based Techniques: Region-based Image segmentation uses three strategies.

Region Growth: At the commencement of pixel-specific region growth algorithms, growth depends on connection with surrounding cells based on grayscale intensity, shape size, or set color thresholds. The planting point and similarity criteria represent regional segmentation. The

algorithm's adaptability depends on its assimilated statistical information and past knowledge [49].

Split-Region Strategy: Second, Image segmentation using split join. Said approach uses a four-tree information representation to divide an Image into four quadrants while considering the original image's non-uniformity. If nearby Images are divided uniformly, they converge into one. Mergers end the cycle. This strategy reduces high-recurrence artifacts by selecting seeding points using breast and cystic mass statistics [50]. These algorithms rely on the image's intensity.

Watershed Approach: Watershed Approach is another region-based method. Image as a top surface. Low-intensity pixels appear as valleys, whereas high-intensity pixels as hills or peaks. If we have water-source seeds, the algorithm fills valleys from local minima. Each seed colors differently with water. These seeds flood local minima to raise overall water. When different-colored waters meet, a barrier prevents them from combining [51]. The obstacle met during filling with water and creating reefs signifies segmentation. This method combines gradient and grayscale information—local catchment area water buildup. Grayscale Images are clones of water pearls, and basins are parts of the image with comparable pixels. Rain and floods are the two major watershed algorithmic processes. The rain algorithm's neighborhood approach finds the image's minimum near the lower district's hallmark. Unmarked pixels have a fictional water drop. The drop moves, gaining significance until it hits a specified pixel. In floods, each bottom quarter keeps theoretical pixels. An ocean of pixels absorbs the lowest pixel's vicinity. Extra pixels are deleted from a pixel flow, making the last one quicker [52]. Images with poor segmentation limitations are not suitable. Standard watershed approaches can over-segment images with noise or low signal-to-noise ratio. An adequate filtering strategy can reduce excessive segmentation. Power watershed algorithms overcome the difficulties of standard watershed algorithms [53]. It combines neural network classification with watershed morphological segmentation to construct breast tumor outlines from ultrasound images. The new stochastic basin version increases contour accuracy. Training adjusts technique parameters. The results reveal a robust technique for automated liver segmentation compared to previous approaches. Marker-driven watershed algorithms and signature combinations removed brain tumors [53].

IV. FEATURE EXTRACTION

When it comes to medical imaging, it is pretty challenging to determine which feature extraction method is the most effective. This remains a general method for extracting features that may be applied to subscribers. When considered as a whole, these depend on the many applications the writers have been working on. How the perpetrator was performed, the data's availability, the data, the dimensionality of the data, the data labeling, and whether or not the method was supervised or unsupervised are the essential elements that need to be taken into consideration when selecting a reliable feature extraction method [55]. A researcher has the option of selecting a suitable feature extraction strategy from among the numerous accessible feature extraction algorithms, which are organized around these processes. An overview of the possible feature extraction methods is presented in the following Figure 4. In the following subsections, the writers provide a concise discussion of each of these methods [56].

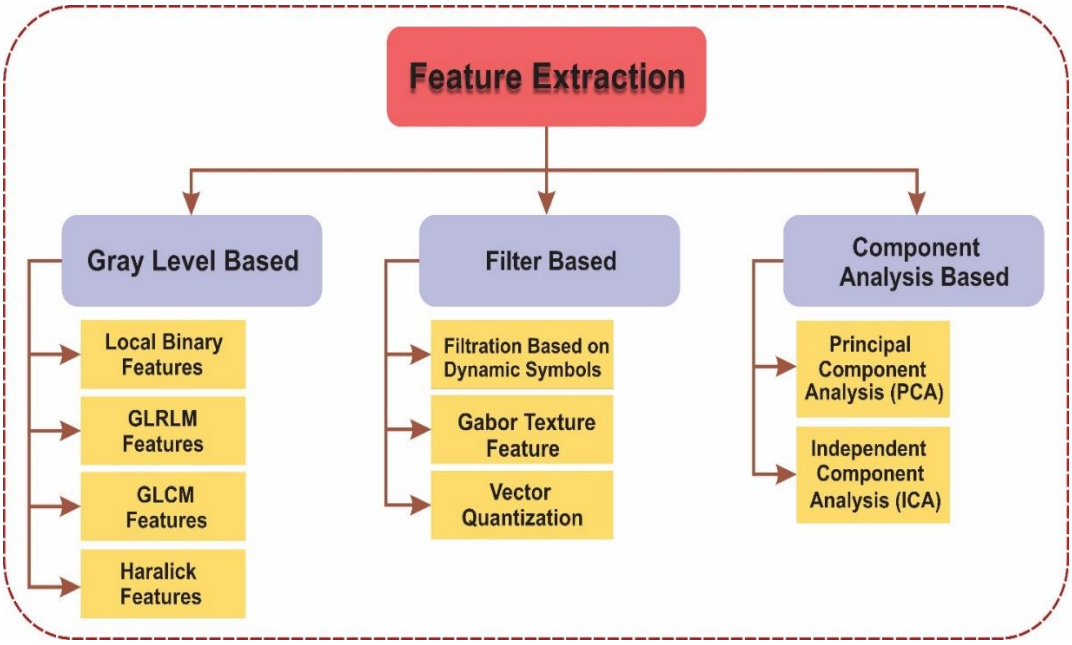


Fig 4 A General Overview of Different Types of Feature Extraction.

A. Local Binary Features

The Local Binary Feature, often known as the LBF, is an example of a storyline fulfilled in mind. The authors' behavior was unchanging about the consistent shifts in grayscale Images. In addition, the writers are pretty swift with their calculations. On the other hand, the writers cannot differentiate between a variety of micro-contours, such as constant bounds, points, and zones. The only information held up by the LBF procedure was the personalizing of the evidence around a neighborhood plot in the immediate area [57]. At the beginning of LBF, the radius is discussed with the surrounding community after some thought and consideration. The LBF operator, seen above, simulates a binary code that determines the local texture pattern when applied to a nearby collection of pixels. Using the gray value of the neighborhood midpoint as a threshold, the binary code may be determined via dithering. In addition, the binary code is reset to a fractional integer that stands in for the LBF code. LBF is a technique that uses grayscale invariant texture information as a result [58-60].

B. Grayscale Run Length Features

Indicating strings of symbols inside an array of Images may be done using the Grayscale Run Length (GLLRM) method encoding [61]. In this instance, the gray level stroke comprises a sequence of successive collinear pixels with the same gray level. The length of a stroke is measured in pixels and looks like this: 0000111100111. In order to extract the features of the texture, this method employs the usage of a stroke length array. The thickness of a texture can also be limited in a specific direction by the stroke length statistics [61]. The GLRM is defined by selecting the direction and counting the incidence of tracks' overall gray levels and lengths traveling in the same direction.

C. Gray-Level Co-Occurrence Matrix Features

The gray-level co-occurrence matrix, abbreviated as GLCM, is a technique for feature extraction that focuses primarily on statistical methodology. Examine the co-occurrence matrix of the gray level or the spatial dependency matrix of the gray level to get an idea of how the pixels are related to one another spatially [62]. The pixel of interest is defined as the spatial connection inside the gray level co-occurrence matrix. Pixels that are horizontally next to one another might create an additional spatial link between two other pixels. It is time to compute the elements of the next gray level co-occurrence matrix. Described, it is the total number of times a pixel's value appears inside the specified spatial connection between a pixel and the input Image [62].

D. Haralick Features

Aspects of the Haralick plot center attention on the co-occurrence matrix produced by the Images. It is necessary to get rid of the co-occurrence matrices to reveal the spatial dependency of the gray level laterally as angular, vertical, and horizontal peaks linkages in the Images [63]. By collapsing the co-occurrence matrix, various characteristics that vary the texture may be produced. Haralick predicted 13 consequent characteristics of the co-occurrence matrix, such as the second angular momentum, contrast, correlation, sum of squares, inverse difference moment, mean of sum, the variance of sum, the entropy of sum, entropy, the variance of difference, entropy of difference, measures of correlation information, and maximum correlation coefficient. Haralick also mentioned the maximum correlation coefficient [63].

E. Filtration Based on Dynamic Symbols

After processing the time series data, the Dynamic Symbolic Filtering (SDF) algorithm removes low-dimensional properties [64]. Wavelet transformation signal preprocessing is used in the dynamic symbolic filtering algorithm to ease the temporal frequency localization and noise elimination process. The existence of the wavelet transform necessitates the change of additional strategic restrictions, including the wavelet base and the collection of scales. Additionally, SDF is a method for extracting space-time features from stationary time series data. This may be done by using the data to create a stationary time series. Identifying abnormalities in complicated dynamic systems at an early stage is arguably the most valuable aspect of this method [64].

F. Gabor Texture Features

The Gabor filter is the foundation of the Gabor feature extraction methodology [65]. These filters are connected to the Gabor wavelets, and their development is considered the human visual system. In addition, it was found that the writers were particularly well suited for studying texture. It is a linear filter that was fundamentally designed for edge detection. Gabor texture features are nothing more than a collection of wavelets. After then, every wavelet acquires vitality in a particular manner and at a specific incidence. Once you have collected the power distributions, you can extract the texture function from the Images [65].

G. Vector Quantization

When the authors have labeled the input data, a supervised learning approach known as Learning Vector Quantization (LVQ) may be used because it is based on vector quantization [66]. Vectors that refer to supervised learning are given a contemporary makeover using the steepest descent method and are employed to minimize the cost function. It generally employs class information to reposition vectors to increase the classification quality of decision areas. This is done in order to maximize accuracy. It begins with LVQ and then moves to two self-organizing maps (SOM) phases. The easiest way to think about it is as a model classification algorithm. The first phase is called feature selection, which involves identifying a very restricted number of characteristics. In most cases, the most critical information from the incoming data is stored in this relatively modest collection of characteristics. The

second phase involves apportioning the feature domains to the various classes [66].

H. An Examination of the Principal Components Analysis (PCA)

Principal Component Analysis is the most excellent approach for extracting features to increase performance when dealing with tiny sizes or situations that need small sizes (PCA) [67]. Additionally, the marking of biased decision restrictions is simple and functional. Alternately, one may refer to it as the Karhunen–Loeve transform (KLT). When studying classification issues, principal component analysis is usually used to reduce the number of dimensions involved. It is generally preferable to identify redundant characteristics that may be deleted from the data sets while analyzing high-dimensional data [68]. This allows for more efficient data management. A classifier based on principal component analysis can be helpful in these situations and can eliminate unneeded features. If all of the characteristics in the feature vector are statistically autonomous, then a person might remove the features that are the least discriminating from the vector. It has been found that several qualities are more dependent on their presence or some unknown underlying variable. One feature might comprise a mix of several different kinds of information to produce a single value. Eliminating this feature would take away more information than is strictly required. In principal component analysis methodology, the features themselves are the primary factors. The authors are orthogonal to one another and create weights that are also orthogonal [58].

I. Independent Component Analysis (ICA)

Independent Component Analysis, often known as ICA, is a method for extracting features in which information is incorporated into accumulating functionality contributing to the whole. These methods are classified under the category of unsupervised learning since they provide a set of vectors that are maximum independent components [69]. This method, which is not regulated by natural forces, is connected to the dispersion of the contribution. On the other hand, it cannot guarantee a satisfactory performance regarding categorization concerns. The feature pattern can comprise separate components of the observed data pattern [70].

V. CONCLUSIONS

The author reviewed numerous well-known methods of Image segmentation and feature extraction, which are relatively recent medical imaging techniques, particularly mammography. Mammography is an effective tool for detecting breast cancer in its earlier stages. If mammograms are utilized without any modifications or improvements, the conclusions reached by a radiologist may cause patients to engage in behaviors that are not essential. In order to hone in on the specifics of the situation, several segmentation strategies from various points of view are utilized. Nevertheless, in light of the shortcomings of the many alternative strategies for segmentation, cluster-based segmentation was deemed superior to the others. Even when using c-mean fuzzy-based segmentation, there are still certain restrictions. In addition, the findings of this research shed light on both the benefits and drawbacks of the methodologies above.

On the other hand, Harlick's texture function is utilized rather frequently for scenarios involving textures. Similarly, Gabor texture features are typically utilized whenever edge detection is required. The writers of this survey are sure that the study contained within it will be valuable to researchers in the future when it comes to segmenting medical images and obtaining characteristics. Clustering is widely recognized as the most effective approach for dividing data sets, and it may be implemented in various settings. A significant number of academics are coming up with ideas for brand new algorithms and even hybridizations of specific already existing algorithms. The Harlick and GLMC texture features are used extensively in various research projects, bringing us to the next part, feature extraction.

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